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Economic Impact Of Ustekinumab In The Treatment Of Psoriasis: A Targeted Literature Review

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Introduction & Objective

Ustekinumab, an interleukin (IL)-12/23 inhibitor biologic, is approved to treat various auto-immune and inflammatory conditions, including psoriasis (PsO). Clinical and real-world evidence has demonstrated its efficacy and safety among patients.

However, its **high cost** may result in reimbursement restrictions that impact overall access and use.

The **objective** of this literature review is to understand the economic impact of ustekinumab's reference product in the treatment of patients with PsO.

Methods

A literature search was conducted in MEDLINE for health-economic modelling publications that evaluate the economic impact of ustekinumab.

The following terms were searched from January 1st, 2018, to March 1st, 2024. Only studies published in English were eligible for inclusion.

Search terms:

- Budget impact model (BIM)
- Cost-effect*
- Cost-utility analysis (CUA)
- Cost per responder (CPR)
- Economic model
- AND PsO

Results

A total of **56** hits were screened for inclusion, resulting in **17** relevant studies that explored the economic impact of ustekinumab in PsO.

Among the cost-effectiveness models published for ustekinumab, the time horizons varied between one year to a lifetime horizon.[1-8] A majority of cost-effectiveness models used a 10-year time horizon (n=4).[2,4,6,7]

Figure 1: Distribution of study types

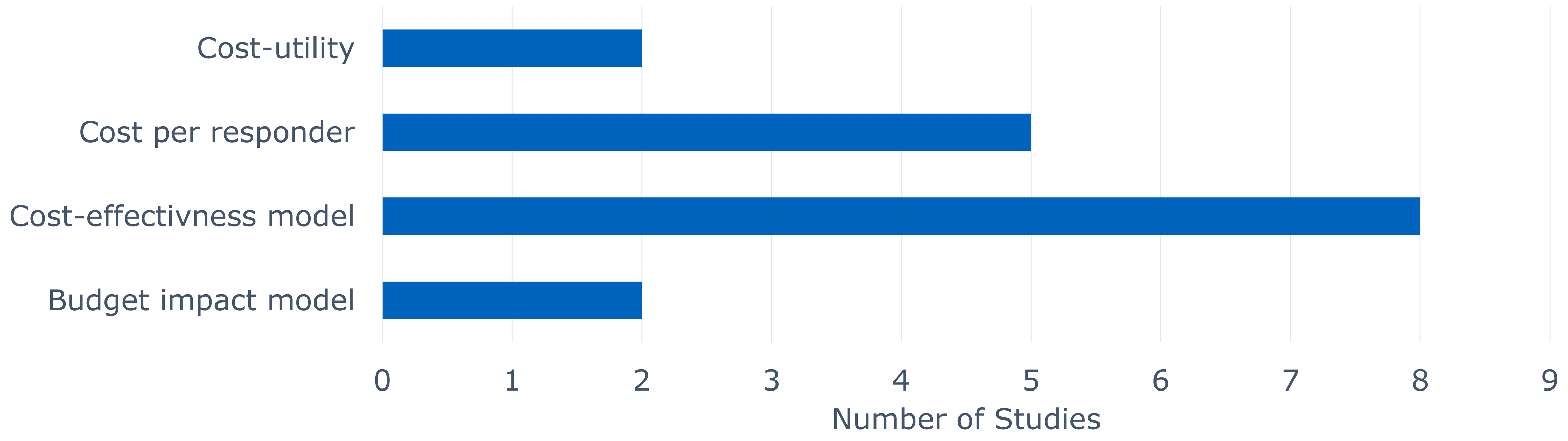

Figure 2: Distribution of pharmacoeconomic analyses, by type and region

	Cost-utility	Cost per responder	Cost-effectiveness	Budget impact model
Australia	☑			
China			☑	
France		☑*		
Germany		☑*	☑	
Italy		☑		
Japan			☑	
Netherlands	☑			
Philippines			☑	☑
Spain		☑		
USA		☑	☑	☑

Results (continued)

Of the studies included, two from the Netherlands and Spain found that ustekinumab was an optimal choice for treatment based on the **positive economic benefit** and high efficiency.[9-10] In the CUA from the Dutch perspective, sequences that **begun** with **ustekinumab** were associated with the **lowest costs** and highest quality-adjusted life years gained.[9]

However, **recent** cost per responder analyses in Italy, France and Germany found ustekinumab was associated with the **highest cost per responder** compared to other biologics.[11-14]

All cost-effectiveness analyses demonstrated that ustekinumab was not a cost-effective therapy at the current willingless to pay for a variety of healthcare settings.[1-8]

The analysis from the Philippine payer perspective noted that the effectiveness of ustekinumab was highly similar to the most cost-effective treatments; however, it was associated with **higher unit drug costs**, which **negatively** affected its incremental cost-effectiveness ratio.[5]

Notably, the included studies did not test the possible cost-effectiveness of an ustekinumab biosimilar which would be expected to reduce the cost associated with treatment, and positively influence economic metrics including budget impact, cost-effectiveness profile, and willingness-to-pay.

Conclusion

This literature review demonstrated that although ustekinumab's reference product is effective in treating patients with PsO, its use may not provide an economic benefit.

As the economic burden for PsO is rising globally, biosimilars would help to address a clear unmet need with regards to the cost of reference product ustekinumab, allowing an effective option to be provided at a lower cost, which would be expected to positively impact healthcare resource utilization and/or patient access.

Conflict of Interest: SR and KS are employees of Fresenius Kabi, a manufacturer of biosimilars. KC and MAG are employees of EVERSANA®, a company that receives consulting fees from Fresenius Kabi.

Footnote: *Nyholm et al. 2023 was conducted from the French and German payer perspective. Abbreviations: CUA = cost-utility analysis; PsO = plaque psoriasis; USA = United States of America.

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